

Balanced rod implies object P must have a smaller weight and hence smaller mass.

7 D Take moments about pivot,

$$M(4.0 \cos 50^\circ) = 45 (7.0 + 4.0 \cos 50^\circ) \Rightarrow M = 170 \text{ N}$$

8 D GPE gained = $mgh = mg \sin \alpha \cdot s$